

We include external scalar and pseudoscalar source fields $\mathcal{S}$ and $\mathcal{P}$, as well as scalar and pseudoscalar diquark source fields $\mathcal{Q}$ and $\mathcal{R}$ ^{#6} as

$$\mathcal{L}_{\text{ext-scalar}} = \bar{\psi}(\mathcal{S} + i\mathcal{P})\psi + \frac{1}{2}[\psi^T C(\gamma_5 \mathcal{Q} + i\mathcal{R})i\tau_2\psi + (\text{h.c.})]. \quad (\text{A.4})$$

The vector and axial-vector external gauge fields $\mathcal{V}^\mu$ and $\mathcal{A}^\mu$ as well as diquark external gauge fields carrying $J^P = 1^+$ and 1^- ($\mathcal{B}^\mu$ and $\mathcal{D}^\mu$) are included as

$$\mathcal{L}_{\text{ext-vector}} = \bar{\psi}\gamma_\mu(\mathcal{V}^\mu - \gamma_5 \mathcal{A}^\mu)\psi + \frac{1}{2}[\psi^T C\gamma_\mu(\mathcal{B}^\mu - \gamma_5 \mathcal{D}^\mu)i\tau_2\psi + (\text{h.c.})]. \quad (\text{A.5})$$

These external fields satisfy the following conditions:

$$\begin{aligned} \mathcal{S}^\dagger &= \mathcal{S}, \quad \mathcal{P}^\dagger = \mathcal{P}, \quad (\mathcal{V}^\mu)^\dagger = \mathcal{V}^\mu, \quad (\mathcal{A}^\mu)^\dagger = \mathcal{A}^\mu, \\ \mathcal{Q}^T &= -\mathcal{Q}, \quad \mathcal{R}^T = -\mathcal{R}, \quad (\mathcal{B}^\mu)^T = \mathcal{B}^\mu, \quad (\mathcal{D}^\mu)^T = -\mathcal{D}^\mu. \end{aligned} \quad (\text{A.6})$$

Now, the total Lagrangian is given by

$$\mathcal{L}_{\text{QCD}} = \mathcal{L}_{\text{QCD}}^0 + \mathcal{L}_{\text{ext-scalar}} + \mathcal{L}_{\text{ext-vector}}. \quad (\text{A.7})$$

It is convenient to introduce two-component spinors q_L and q_R in such a way that the four-component spinor ψ is expressed as $\psi^i = \begin{pmatrix} q_L^i \\ q_R^i \end{pmatrix}$, where i denotes the flavor index. Then, the kinetic term of quarks is rewritten as

$$\int d^4x i\bar{\psi}_i\gamma_\nu D^\nu\psi^i = \int d^4x (iq_{L,i}^\dagger\sigma_\nu D^\nu q_L^i + iq_{R,i}^\dagger\bar{\sigma}_\nu D^\nu q_R^i), \quad (\text{A.8})$$

where we use the following form of the gamma matrices:

$$\gamma_\nu = \begin{pmatrix} \mathbf{0} & \bar{\sigma}_\nu \\ \sigma_\nu & \mathbf{0} \end{pmatrix} \equiv \begin{pmatrix} \mathbf{0} & (\mathbf{1}, \sigma_j) \\ (\mathbf{1}, -\sigma_j) & \mathbf{0} \end{pmatrix}. \quad (\text{A.9})$$

From the form given in Eq.(A.8) the existence of the $SU(2N_f)$ flavor symmetry is seen as follows: Since the fundamental representation of $SU(2)_{\text{color}}$ as well as that of $SU(2)_{\text{spin}}$ is the pseudoreal representation, a combination of $\sigma_2\tau_2 q_R^*$ has the same transformation property

^{#6} $\mathcal{Q}$ and $\mathcal{R}$ are color-singlet states in two-color QCD. This can be seen from the existence of $i\tau_2 = \epsilon$ and the charge-conjugation matrix $C \equiv i\gamma_2\gamma_0$ in Eq. (A.4).